

What's New: PythonScript Agent

Adroit 11 Agent Server | Release highlight

Executive Summary

Adroit 11 introduces the PythonScript Agent — a new Agent Server agent that executes Python scripts as part of your SCADA automation strategy. Scripts run outside the Agent Server process to protect server stability, while the Adroit OLE Python package (adroit-py) gives scripts full read/write access to tags, slot information, and historical data through Adroit's OLE Automation interface.

Whether you need event-driven calculations, custom integrations, or advanced data processing at the Agent Server layer, the PythonScript Agent brings Python's ecosystem to Adroit without compromising reliability.

Property	Value
General	
Agent description	Sample PythonScript agent
Execution	
enable	True
synchronous	True
runNow	False
triggerList	2 rows, 5 cols ...
pythonPath	C:\Users\vnikel\AppData\Local\Programs\Python\Python314\python.exe
Script file	
scriptFn	\$py
parameters	Sleep 1 5
editorPath	C:\Program Files\TextPad 8\TextPad.exe
Information	
info	Process finished with error code: 1
runTime	32
Status	
busy	False
complete	False
error	True

Agent description
Description of agent

Figure 1: PythonScript Agent Configuration

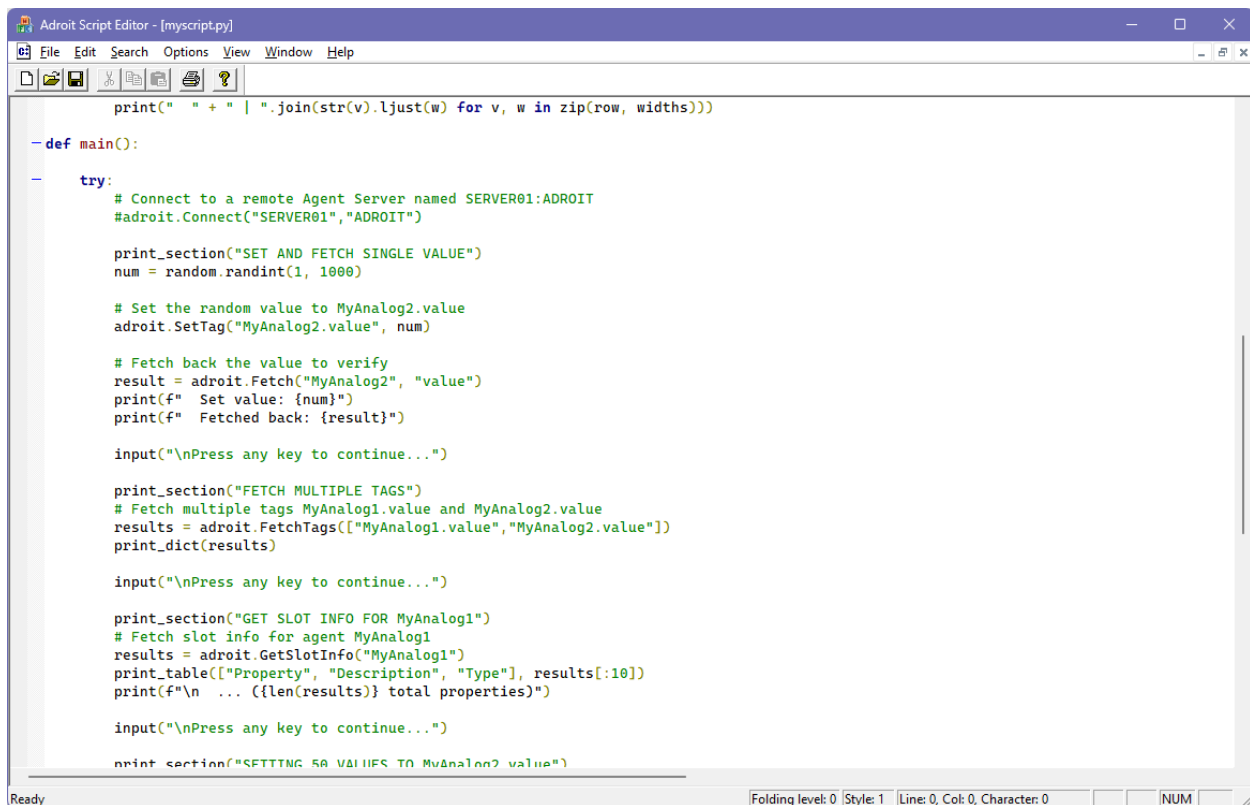


Figure 2: New Python Script Editor (adrscripedit.exe)

Key Features at a Glance

Feature	What it does	Why it matters
PythonScript Agent	New Agent Server agent (T_PythonScript.dll) that launches Python scripts on demand or via tag triggers	Brings Python automation to the Agent Server layer alongside existing Adroit agents
New Python Script Editor	New Python script editor powered by the Scintilla, featuring syntax highlighting, code folding, and an enhanced editing experience for faster, more efficient script development.	It makes maintaining Python scripts significantly easier and increases productivity and improve the overall scripting experience.
Process isolation	Scripts execute in a separate python.exe process, outside the Agent Server	Protects Agent Server stability — a script failure cannot crash the server

Adroit OLE Python (adroit-py)	Python package for Fetch, SetTag, FetchTags, FetchChanges, FetchValues, GetSlotInfo, and remote Connect	Scripts interact with live SCADA data using a simple, purpose-built API
Trigger tags	Configure a list of trigger tags with enable/disable, value change, rising edge, or falling edge conditions	Run scripts automatically in response to process events
Sync / async execution	Run synchronously (wait for completion) or asynchronously (fire-and-forget); always on a dedicated thread	Flexibility for short calculations vs. long-running integrations without blocking other agents
Script parameters	Pass optional command-line parameters to scripts via the parameters slot	Reuse one script for multiple scenarios
Configurable Python path	Point to Adroit-installed Python or your own python.exe installation	Works with existing Python environments or Adroit's bundled setup
Status feedback	Busy, Complete, and Error status bits plus info and runTime slots	Operators and logic can monitor script execution and detect failures

Feature Highlights

Python automation at the Agent Server

The PythonScript Agent is a first-class Agent Server agent. Add it to your project like any other agent, configure the script file, and execute it on demand via the runNow slot or automatically through configured trigger tags.

Customer benefit: Teams can implement custom logic, calculations, and integrations at the SCADA layer using Python — without custom DLL development.

Safe, isolated script execution

Python scripts run in a separate process (python.exe) with no visible console window. The Agent Server launches and monitors the process while continuing to serve other agents normally.

Customer benefit: Script errors are contained. A non-zero exit code sets the agent's Error status and reports the exit code in the info slot — the Agent Server keeps running.

Full SCADA integration via Adroit OLE Python

The adroit-py package provides the AdroitOLE wrapper, enabling scripts to fetch and set tag values, retrieve slot information, query historical changes and values, and connect to remote Agent Servers. Adroit 11 can install Python, pywin32, and adroit-py automatically; existing Python installations are supported with pip install.

Customer benefit: Leverage Python's rich ecosystem (pandas, requests, ML libraries, etc.) while reading and writing live process data.

Event-driven execution with trigger tags

Configure one or more trigger tags per agent. Each trigger can be individually enabled or disabled and set to fire on any value change, rising edge ($0 \rightarrow n$), or falling edge ($n \rightarrow 0$).

Customer benefit: Run Python logic in direct response to process events — alarm conditions, setpoint changes, batch phase transitions, and more.

Flexible execution modes

Scripts always launch on a dedicated thread. In synchronous mode, the agent waits for the Python process to finish before pulsing Complete. In asynchronous mode, the agent returns immediately while the script continues in the background.

Customer benefit: Choose the right execution model for each use case — quick tag calculations vs. long-running data exports or API calls.

New Python Script Editor

The new Python Script Editor improves productivity by making scripts easier to read, write, and maintain. Syntax highlighting helps users quickly identify code structure and potential errors, while code folding simplifies navigation through large scripts.

By default, Adroit opens scripts in the new Python Script Editor. If you prefer, you can use your own script editor by specifying the full path to its executable.

Customer benefit: Faster development, easier troubleshooting, and a more efficient scripting workflow.

Who Benefits

Audience	Value
SCADA engineers and integrators	Implement custom calculations, data transformations, and integrations without C++, VB Script or JScript development

Automation developers	Use Python skills and libraries directly at the Agent Server layer
Operations teams	Monitor script execution through standard agent status bits and slots
IT / OT integration teams	Bridge SCADA data to enterprise systems via Python REST, database, and messaging libraries
Organizations adopting Python	Standardize on one language across Smart UI scripting and Agent Server automation

Typical Use Cases

- Custom calculations and engineering unit conversions triggered by tag changes
- Data aggregation and KPI computation at the Agent Server
- REST API and web service integration with external systems
- Database read/write operations for MES, ERP, or CMMS connectivity
- Batch and recipe logic extensions using Python's expressiveness
- Historical data analysis and reporting triggered on schedule or event
- Alarm enrichment and notification routing via custom Python logic
- Proof-of-concept and rapid prototyping before formal agent development

Technical Scope (internal reference)

- Product: Adroit 11 Agent Server
- Agent DLL: T_PythonScript.dll
- Script Editor: adrscripedit.exe
- Python packages: pywin32, adroit-py (Adroit OLE Python)
- Default Python path: C:\Program Files\Adroit Technologies\Python\Python.exe
- Key slots: scriptFn, pythonPath, parameters, synchronous, runNow, triggerList, runTime, info
- Status bits: busy, complete, error
- Future: embedded mode (in-process Python execution) planned for a later release
- Not supported: script scheduling (use trigger tags or external Scheduler agent), realtime catchup

Document Information

Based on: PythonScript Agent URS/FDS (PythonScript Agent.docx) and Adroit 11 Agent Server help topics

For detailed end-user instructions, refer to the Adroit 11 help: [The PythonScript Agent](#), [Configuring PythonScript Agents](#), and [PythonScript Agent Slot Details](#)